Flow Descriptions and Flow Lines

Dr. Huan Phan
Assistant Professor, Robotics and Mechatronics Engineering
Randolph College

Fluid Mechanics Lecture Notes

Learning Objectives

After this lecture, students should be able to:

- distinguish between Eulerian and Lagrangian descriptions of fluid motion;
- define and interpret streamlines, pathlines, and streaklines;
- derive and solve the differential equation of a streamline;
- formulate the equations that describe a fluid-particle pathline;
- explain when the three types of flow lines coincide; and
- calculate and compare flow lines in a simple unsteady flow.

1 Why Flow Descriptions Matter

A fluid flow is described by a velocity field

$$\boldsymbol{V}(x, y, z, t) = u(x, y, z, t)\,\boldsymbol{i} + v(x, y, z, t)\,\boldsymbol{j} + w(x, y, z, t)\,\boldsymbol{k}, \tag{1}$$

where u, v, and w are the velocity components in the x, y, and z directions. Although this field contains the complete kinematic information, a picture made from curves is often easier to interpret. Flow lines reveal the direction of motion, the history of a particle, or the location of particles that passed through a common source.

The meaning of the picture depends on how the curves are constructed. A streamline, a pathline, and a streakline are not interchangeable definitions.

2 Two Descriptions of Fluid Motion

2.1 Eulerian Description

In the Eulerian approach, we select locations in space and ask how the flow properties at those locations change with time. The independent variables are position and time:

$$\boldsymbol{V} = \boldsymbol{V}(\boldsymbol{x}, t) = \boldsymbol{V}(x, y, z, t). \tag{2}$$

For example, a velocity probe fixed inside a wind tunnel gives an Eulerian measurement. It reports the velocity passing the probe, but it does not follow one particular fluid particle.

2.2 Lagrangian Description

In the Lagrangian approach, each particle is labeled by its initial position $\boldsymbol{a} = (a, b, c)$. Its position at time t is

$$\boldsymbol{X} = \boldsymbol{X}(\boldsymbol{a}, t). \tag{3}$$

The velocity of that particle is the time derivative of its position:

$$\frac{d\boldsymbol{X}}{dt} = \boldsymbol{V}\big(\boldsymbol{X}(\boldsymbol{a}, t), t\big). \tag{4}$$

Small, neutrally buoyant tracer particles are used experimentally to approximate this description. Their inertia must be small enough that they faithfully follow the surrounding fluid.

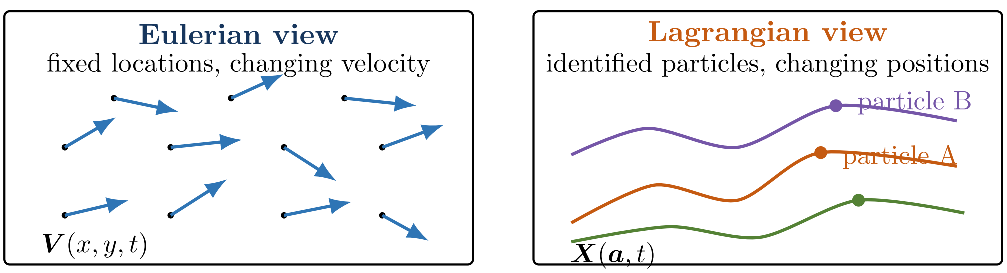

Figure 1: The Eulerian description observes the flow at fixed spatial points; the Lagrangian description follows identified fluid particles.

2.3 Connecting the Two Descriptions

Equation (4) connects the Eulerian velocity field to a Lagrangian particle trajectory. It also leads to the acceleration of a fluid particle:

$$\begin{aligned}
\boldsymbol{a} &= \frac{d}{dt}\boldsymbol{V}\big(\boldsymbol{X}(t), t\big) && (5)\\
&= \frac{\partial \boldsymbol{V}}{\partial t} + u\frac{\partial \boldsymbol{V}}{\partial x} + v\frac{\partial \boldsymbol{V}}{\partial y} + w\frac{\partial \boldsymbol{V}}{\partial z} && (6)\\
&= \frac{D\boldsymbol{V}}{Dt}. && (7)
\end{aligned}$$

The first term is the *local acceleration*; the remaining terms form the *convective acceleration.* This material derivative allows Newton's second law to be applied to a moving fluid particle while the flow is stored as an Eulerian field.

Table 1: Eulerian and Lagrangian descriptions

	Eulerian	Lagrangian
Question	What is happening at this location?	What happens to this particle?
Independent variables	Position $\boldsymbol{x}$ and time t	Particle label $\boldsymbol{a}$ and time t
Main quantity	$\boldsymbol{V}(\boldsymbol{x}, t)$	$\boldsymbol{X}(\boldsymbol{a}, t)$
Typical measurement	Fixed velocity probe	Tracer-particle tracking
Common use	Describing a complete flow field	Particle motion and Newton's second law

3 Streamlines

A **streamline** is a curve that is everywhere tangent to the *instantaneous* velocity field. It is a snapshot of flow direction at one chosen time $t = t_*$.

Let an infinitesimal displacement along the streamline be

$$d\boldsymbol{r} = dx\,\boldsymbol{i} + dy\,\boldsymbol{j} + dz\,\boldsymbol{k}. \tag{8}$$

Because $d\boldsymbol{r}$ and $\boldsymbol{V}$ are parallel,

$$d\boldsymbol{r} \times \boldsymbol{V} = \boldsymbol{0}. \tag{9}$$

Therefore, the components are proportional:

$$\boxed{\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}} \qquad \text{at a fixed time } t = t_*. \tag{10}$$

For a two-dimensional flow, the usual streamline equation is

$$\boxed{\left.\frac{dy}{dx}\right|_{t=t_*} = \frac{v(x, y, t_*)}{u(x, y, t_*)}.} \tag{11}$$

The integration constant identifies which member of the family of streamlines is being considered. A point through which the streamline passes supplies the required condition.

Key Idea

When solving for a streamline, freeze the time at $t = t_*$. Integrate with respect to position; do not allow t to vary along the curve. A streamline is an instantaneous geometric curve, not automatically the trajectory of one particle.

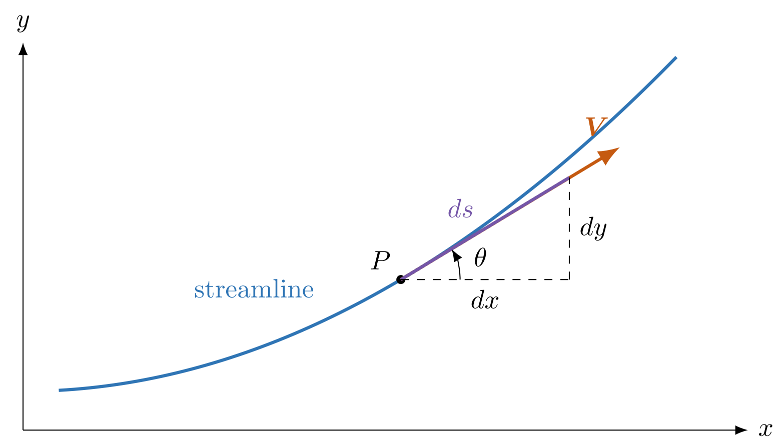

Figure 2: On a streamline, the differential displacement $d\boldsymbol{r}$ and the local velocity $\boldsymbol{V}$ are parallel. Thus $dy/dx = v/u$ in two dimensions.

4 Pathlines

A **pathline** is the actual trajectory followed by one identified fluid particle. If the particle is at (x_0, y_0, z_0) at time t_0, its pathline is found from

$$\boxed{\frac{dx}{dt} = u(x, y, z, t), \qquad \frac{dy}{dt} = v(x, y, z, t), \qquad \frac{dz}{dt} = w(x, y, z, t)} \tag{12}$$

with

$$x(t_0) = x_0, \qquad y(t_0) = y_0, \qquad z(t_0) = z_0. \tag{13}$$

The result is often most naturally written as parametric equations $x = x(t)$, $y = y(t)$, and $z = z(t)$. If desired, the parameter t can then be eliminated to produce a geometric equation.

Notice an important substitution: the Eulerian field must be evaluated at the particle's *current* position. Therefore, x, y, and z inside the right-hand side of Eq. (12) are themselves functions of time.

5 Streaklines

A **streakline** is the curve formed at an observation time $t = T$ by all particles that previously passed through the same fixed spatial point. A continuous stream of smoke from a small port or dye from a needle produces a streakline.

Suppose particles pass through a source location $\boldsymbol{x}_s$ at different release times τ. For every value of τ, solve the particle equation

$$\frac{d\boldsymbol{X}}{dt} = \boldsymbol{V}(\boldsymbol{X}, t), \qquad \boldsymbol{X}(\tau; \tau) = \boldsymbol{x}_s. \tag{14}$$

At the observation time T, the streakline is the locus

$$\boxed{\boldsymbol{x}_{\text{streak}}(\tau; T) = \boldsymbol{X}(T; \tau), \qquad \tau \le T.} \tag{15}$$

One pathline follows one particle through many times. One streakline connects many particles at one observation time.

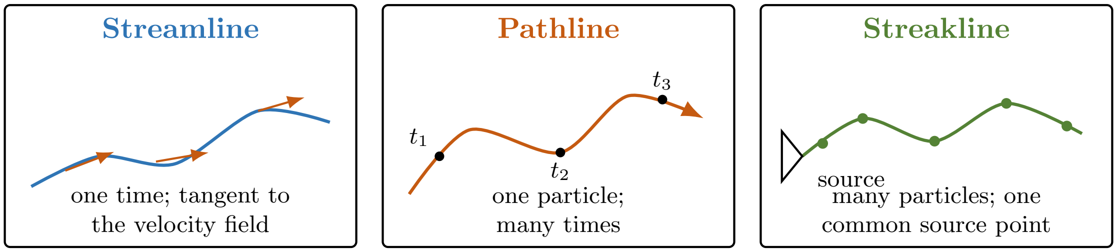

Figure 3: The three flow-line definitions answer different questions.

Table 2: Comparison of the three flow lines

	Streamline	Pathline	Streakline
Connects	Points tangent to $\boldsymbol{V}$ at one instant	Positions of one particle over time	Particles that passed through one fixed point
Time meaning	Time is frozen	Time evolves	One observation time, many release times
How to observe	Instantaneous direction field	Track one tracer	Continuous smoke or dye injection
Main relation	$dx/u = dy/v = dz/w$	$d\boldsymbol{X}/dt = \boldsymbol{V}$	$\boldsymbol{X}(T;\tau)$ for many τ

6 When the Three Flow Lines Coincide

A flow is steady when

$$\frac{\partial \boldsymbol{V}}{\partial t} = \boldsymbol{0}. \tag{16}$$

In a steady flow, the direction field does not change while a particle moves through it. A particle remains on the same streamline, and every particle released from a fixed source follows the same curve. Therefore,

$$\boxed{\text{steady flow: streamline} = \text{pathline} = \text{streakline.}} \tag{17}$$

In an unsteady flow, the instantaneous direction field can rotate or deform while a particle travels. The three curves are then generally different. Special unsteady fields may still produce coincident curves, so steadiness is a sufficient condition, not a necessary condition in every exceptional case.

7 Worked Example: Oscillating Jet

Problem Statement

A small fixed nozzle at the origin produces a jet with constant downstream velocity U_0 while the transverse velocity at the nozzle oscillates. The oscillation signal is convected downstream at speed U_0. Neglect gravity and use the two-dimensional Eulerian field

$$u = U_0, \qquad v = V_0 \sin\left[\omega\left(t - \frac{x}{U_0}\right)\right]. \tag{18}$$

Let

$$U_0 = 2.0\,\mathrm{m/s}, \qquad V_0 = 0.50\,\mathrm{m/s}, \qquad T_p = 4.0\,\mathrm{s}.$$

Find: (a) the streamline through the origin at $t_* = 2.0\,\mathrm{s}$; (b) the pathline of the particle released at $\tau = 0$; (c) the pathline of the particle released at $\tau = 1.0\,\mathrm{s}$; and (d) the streakline at the observation time $T = 2.0\,\mathrm{s}$.

The angular frequency is

$$\omega = \frac{2\pi}{T_p} = \frac{2\pi}{4.0\,\mathrm{s}} = \frac{\pi}{2}\ \mathrm{rad/s}. \tag{19}$$

(a) Streamline at $t_* = 2.0\,\mathrm{s}$

Freeze time and use Eq. (11):

$$\frac{dy}{dx} = \frac{V_0}{U_0}\sin\left[\omega\left(t_* - \frac{x}{U_0}\right)\right]. \tag{20}$$

Integrating with respect to x while keeping t_* constant gives

$$y = \frac{V_0}{\omega}\cos\left[\omega\left(t_* - \frac{x}{U_0}\right)\right] + C. \tag{21}$$

The streamline passes through $(0,0)$, so

$$C = -\frac{V_0}{\omega}\cos(\omega t_*). \tag{22}$$

Thus,

$$y_s(x;t_*) = \frac{V_0}{\omega}\left\{\cos\left[\omega\left(t_* - \frac{x}{U_0}\right)\right] - \cos(\omega t_*)\right\}. \tag{23}$$

At $t_* = 2.0\,\mathrm{s}$,

$$\boxed{y_s(x) = 0.318\left[1 - \cos\left(\frac{\pi x}{4}\right)\right]\ \mathrm{m}} \tag{24}$$

when x is measured in meters.

(b) Pathline of the particle released at $\tau = 0$

For a particle released from $(0, 0)$ at time $t = \tau$, first integrate the horizontal motion:

$$\frac{dx}{dt} = U_0 \quad \Longrightarrow \quad x = U_0(t - \tau). \tag{25}$$

Substitute this result into the transverse velocity:

$$\frac{dy}{dt} = V_0 \sin\left[\omega\left(t - \frac{U_0(t - \tau)}{U_0}\right)\right] \tag{26}$$

$$= V_0 \sin(\omega\tau). \tag{27}$$

The phase experienced by one particle is constant. Integrating from its release time gives

$$y = V_0 \sin(\omega\tau)(t - \tau). \tag{28}$$

Eliminating $t - \tau = x/U_0$,

$$\boxed{y_p = \frac{V_0}{U_0} \sin(\omega\tau)\, x.} \tag{29}$$

Every pathline in this model is straight because $Dv/Dt = \partial v/\partial t + U_0\, \partial v/\partial x = 0$; the local and convective accelerations are nonzero but cancel exactly.

For $\tau = 0$, $\sin(0) = 0$, so

$$\boxed{x = 2t, \qquad y_p = 0.} \tag{30}$$

At $T = 2.0\,\mathrm{s}$, this particle is at $(x, y) = (4.0\,\mathrm{m}, 0)$.

(c) Pathline of the particle released at $\tau = 1.0\,\mathrm{s}$

Because $\omega\tau = (\pi/2)(1) = \pi/2$,

$$x = 2(t - 1), \qquad y = 0.50(t - 1). \tag{31}$$

Therefore,

$$\boxed{y_p = 0.25x.} \tag{32}$$

At $T = 2.0\,\mathrm{s}$, this particle is at $(x, y) = (2.0\,\mathrm{m}, 0.50\,\mathrm{m})$.

(d) Streakline at $T = 2.0\,\mathrm{s}$

At the observation time T, a particle released at time τ is located at

$$x = U_0(T - \tau). \tag{33}$$

Therefore,

$$\tau = T - \frac{x}{U_0}. \tag{34}$$

Substitute this release time into Eq. (29):

$$y_{\text{streak}}(x;T) = \frac{V_0}{U_0} x \sin\left[\omega\left(T - \frac{x}{U_0}\right)\right]. \tag{35}$$

At $T = 2.0\,\text{s}$,

$$\boxed{y_{\text{streak}}(x) = 0.25x \sin\left(\frac{\pi x}{4}\right), \qquad 0 \le x \le 4.0\,\text{m}.} \tag{36}$$

The domain ends at $x = U_0 T = 4.0\,\text{m}$ because the earliest released particle has not traveled farther than that distance.

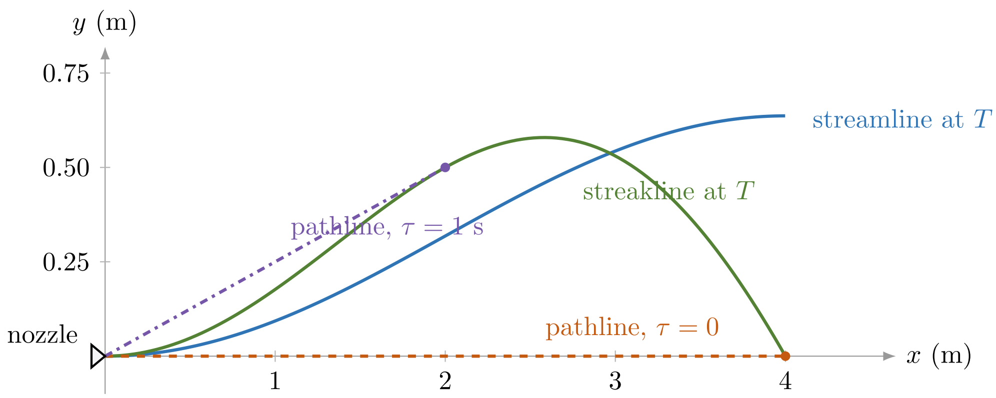

Figure 4: At $T = 2.0\,\text{s}$, the streamline, two particle pathlines, and the streakline are different because the velocity field is unsteady. The colored dots show the two particles' positions at the observation time.

For example, at $x = 2.0\,\text{m}$ the streamline through the origin is at $y_s = 0.318\,\text{m}$, whereas the streakline is at $y_{\text{streak}} = 0.500\,\text{m}$. The curves answer different physical questions, so there is no reason for them to agree in this unsteady flow.

Key Idea

Along a pathline, x is a function of time. In this example, substituting $x = U_0(t - \tau)$ before integrating dy/dt makes the phase constant. Treating x as an independent constant during the pathline integration would give the wrong trajectory.

8 Common Mistakes

1. **Allowing time to vary while calculating a streamline.** A streamline is calculated at one frozen time $t = t_*$. Use Eq. (11) as a spatial differential equation.
2. **Calling every visible dye curve a streamline.** A continuous dye trace is a streakline. It represents a streamline only when the flow is steady or when a special unsteady field makes the curves coincide.
3. **Forgetting that x, y, and z depend on time along a pathline.** Substitute the particle position into the Eulerian velocity field before integrating.

4. **Thinking a streakline follows one particle.** A streakline contains many particles released at different times from one fixed location.
5. **Assuming tracer particles always follow the fluid exactly.** Large or heavy tracers can slip relative to the fluid because of inertia, buoyancy, or settling.

9 In-Class Concept Questions

1. A dye needle continuously releases color into a steady water flow. Which flow line is directly visible, and under what condition can it also be called a streamline?
2. In a spatially uniform flow whose direction changes with time, the streamlines at each instant are straight. Must the pathlines also be straight?
3. In Eq. (11), why is time treated as a parameter, whereas it is the independent variable in Eq. (12)?

Brief answers. (1) A streakline is visible; it is also a streamline in steady flow. (2) No. A particle can follow a curved path as the velocity direction changes. (3) A streamline is an instantaneous spatial snapshot, while a pathline records the time history of one particle.

10 Summary

$$
\begin{aligned}
\text{Eulerian velocity field:} &\quad \boldsymbol{V} = \boldsymbol{V}(x, y, z, t), \\
\text{Lagrangian trajectory:} &\quad \frac{d\boldsymbol{X}}{dt} = \boldsymbol{V}(\boldsymbol{X}, t), \\
\text{Material acceleration:} &\quad \frac{D\boldsymbol{V}}{Dt} = \frac{\partial \boldsymbol{V}}{\partial t} + (\boldsymbol{V} \cdot \nabla)\boldsymbol{V}, \\
\text{Streamline:} &\quad \frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} \quad (t = t_*), \\
\text{Two-dimensional streamline:} &\quad \left.\frac{dy}{dx}\right|_{t=t_*} = \frac{v}{u}, \\
\text{Pathline:} &\quad \frac{dx}{dt} = u, \quad \frac{dy}{dt} = v, \quad \frac{dz}{dt} = w, \\
\text{Streakline at time } T\text{:} &\quad \boldsymbol{x}_{\text{streak}}(\tau; T) = \boldsymbol{X}(T; \tau), \\
\text{Steady flow:} &\quad \text{streamline} = \text{pathline} = \text{streakline}.
\end{aligned}
$$